Message Digest 5 (MD5)

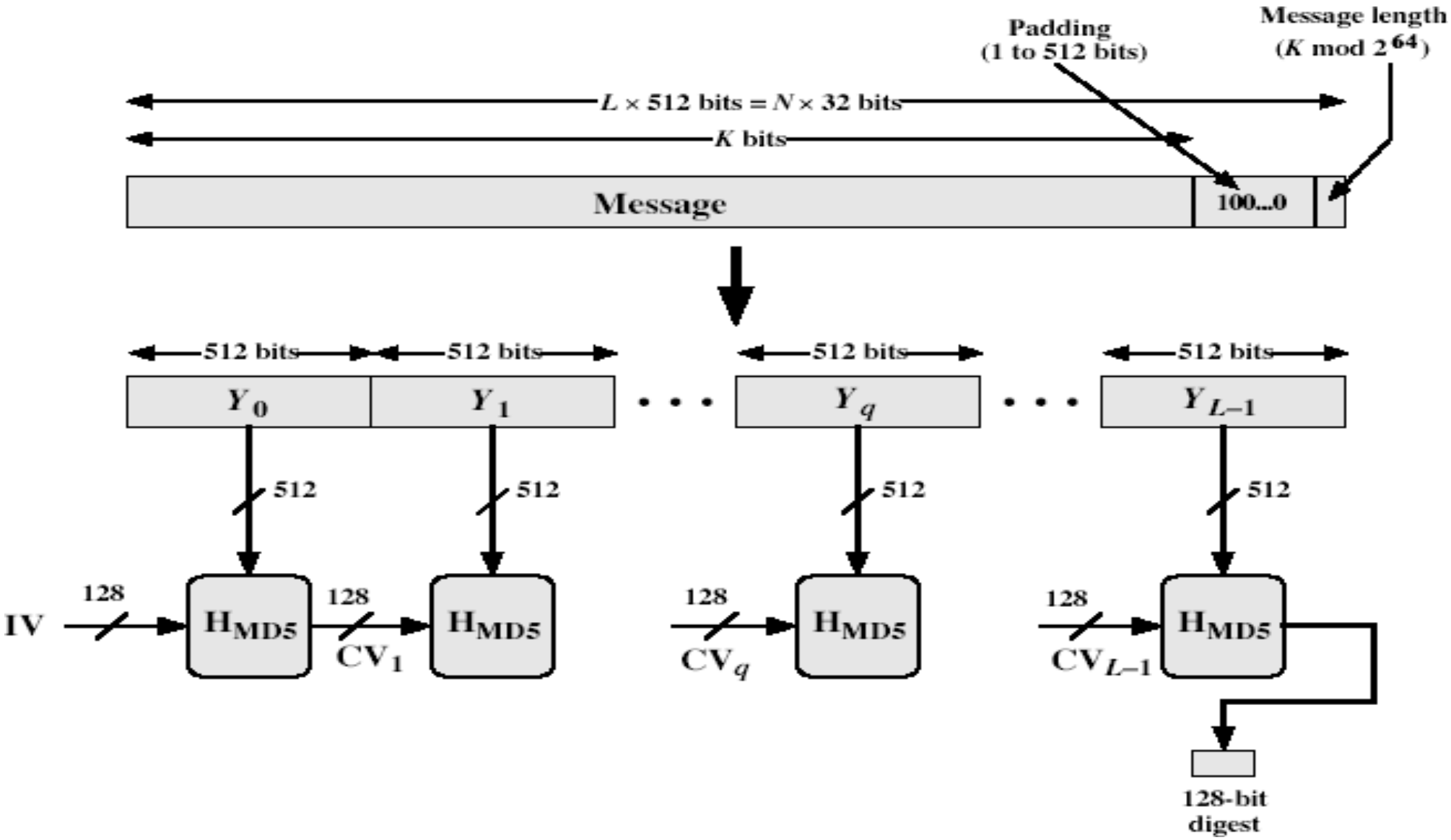
Introduction

- Designed by *Ronald Rivest* (the "*R*" in RSA)
- Latest in a series of MD2, MD4
- Produces a 128-bit hash value
- Until recently was the most widely used hash algorithm
  - in recent times have both brute-force & cryptanalytic concerns
- Specified as Internet standard RFC 1321
- According to RFC 1321, "MD5 message-digest algorithm takes as input a message of arbitrary length and produces as output a 128-bit "message digest" of the input …The MD5 algorithm is intended for digital signature applications, where a large file must be "compressed" in a secure manner before being encrypted with a private (secret) key under a public-key cryptosystem such as RSA."

MD5 Overview

1. Pad message so its length is 448 mod 512
2. Append a 64-bit length value to message
3. Initialise 4-word (128-bit) MD buffer (A,B,C,D)
4. Process message in 16-word (512-bit) blocks:
   - using 4 rounds of 16 bit operations on message block & buffer
   - add output to buffer input to form new buffer value
5. Output hash value is the final buffer value

Structure of MD5



MD5 Implementation Steps

Step1: Append padding bits

- The input message is "padded" (extended) so that its length (in bits) equals to 448 mod 512
- Padding is always performed, even if the length of the message is already 448 mod 512
- Padding is performed as follows: a single "1" bit is appended to the message, and then "0" bits are appended so that the length in bits of the padded message becomes congruent to 448 mod 512
- At least one bit and at most 512 bits are appended

MD5 Implementation Steps … Contd.

Step 2: Append length

- A 64-bit representation of the length of the message is appended to the result of step1
- If the length of the message is greater than 2^{64}, only the low-order 64 bits will be used
- The resulting message (after padding with bits and with L) has a length that is an exact multiple of 512 bits
- The input message will have a length that is an exact multiple of 16 (32-bit) words

MD5 Implementation Steps … Contd.

Step 3: Initialize MD buffer

- A four-word buffer (A, B, C, D) is used to compute the message digest
- Each of A, B, C, D is a 32-bit register
- These registers are initialized to the following values in hexadecimal, low-order bytes first):

  word A: 01 23 45 67

  word B: 89 ab cd ef

  word C: fe dc ba 98

  word D: 76 54 32 10

MD5 Implementation Steps … Contd.

Step 4: Process message in 16-word blocks

- Four functions will be defined such that each function takes an input of three 32-bit words and produces a 32-bit word output

  F (X, Y, Z) = (X and Y) or (not (X) and Z)

  G (X, Y, Z) = (X and Z) or (Y and not (Z))

  H (X, Y, Z) = X xor Y xor Z

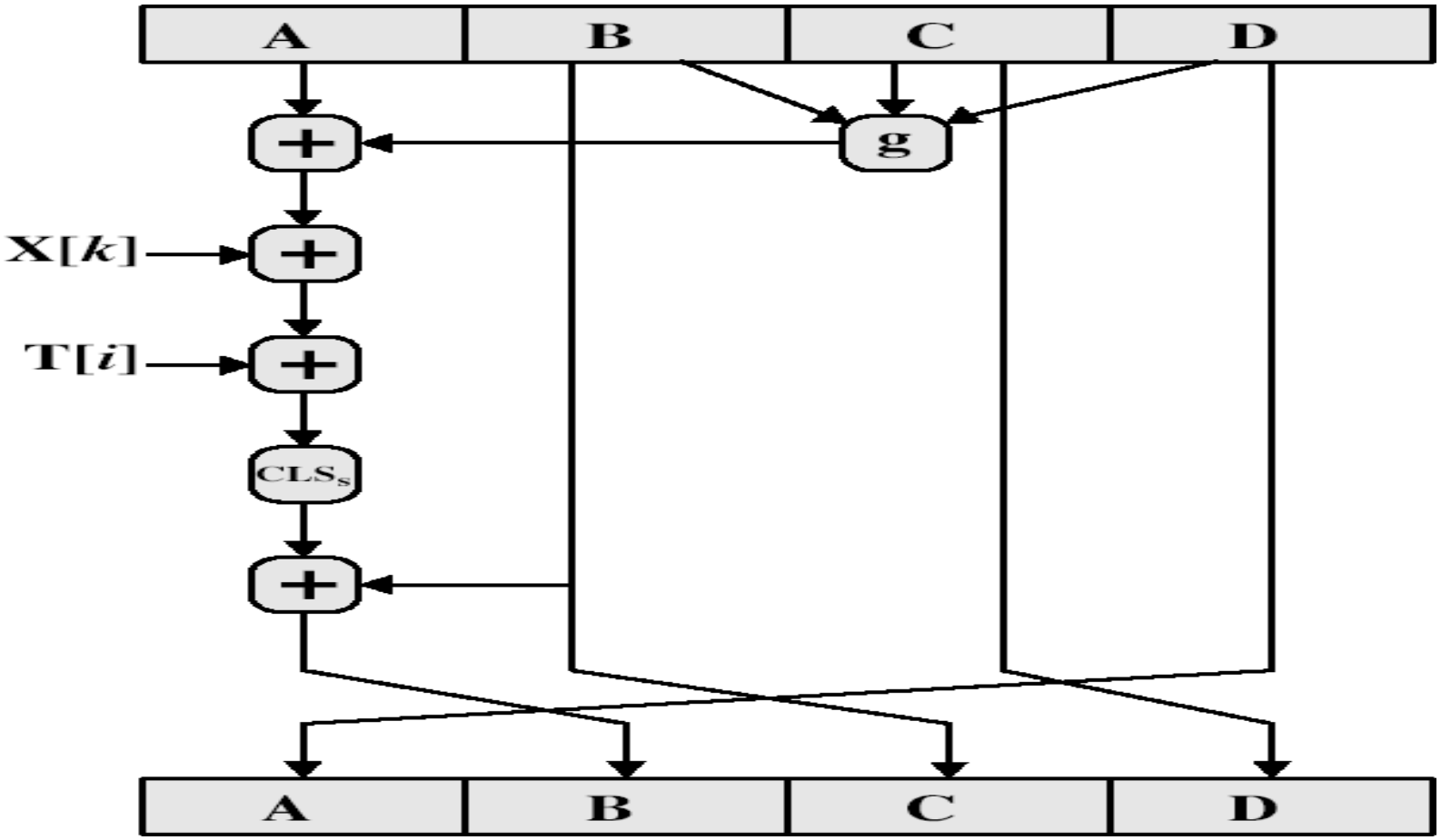
  I (X, Y, Z) = Y xor (X or not (Z))

MD5 Compression Function

- Each round has 16 steps of the form:

  A = B+((A+g(B,C,D)+X[k]+T[i])<<<s)
- A,B,C,D refer to the 4 words of the buffer, but used in varying permutations
  - note this updates 1 word only of the buffer
  - after 16 steps each word is updated 4 times
- where g(B,C,D) is a different nonlinear function in each round (F,G,H,I)
- T[i] is a constant value derived from sin function=abs(sin(i+1))*2^32

MD5 Compression Function … Contd.



Strength of MD5

- MD5 hash is dependent on all message bits
- Rivest claims security is good as can be
- Known attacks are:
  - ✓ Berson 92 attacked any 1 round using differential cryptanalysis (but can't extend)
  - ✓ Boer & Bosselaers 93 found a pseudo collision (again unable to extend)
  - ✓ Dobbertin 96 created collisions on MD compression function (but initial constants prevent exploit)
- MD5 looks vulnerable soon

Practice Questions

Assume that you are a software engineer at a company responsible for managing sensitive customer data. One of your tasks is to implement a secure system for verifying the integrity of files that customers upload to the company's cloud storage and you have decided to use the MD5 hashing algorithm.

a) Illustrate the algorithm to ensure the integrity of the uploaded files.

b) Identify the potential security risks that are associated with using MD5 in this context, and suggest an alternative algorithm you might consider to overcome the security risks.

Mr. Arun is hashing the message **"Information Security"** using the MD5 algorithm.
Explain the steps involved in **one round of MD5 hashing**, including:

- Use of nonlinear functions (F, G, H, I)
- Use of constants and shift operations
- Updating buffer registers

Find the padding length in MD5 for the following message sizes:
a) 1000 bits b) 2047 bits

Hash the message **"Hello"** using MD5 preprocessing steps. Explain:

- Padding
- Message block formation
- Message schedule preparation